

NUMBER OF PINS	
No. of Digital Inputs: Higher the number of sensors (with digital input) you wish to connect, higher the number of pins that you'll need. Higher the number of output devices connected to a single microcontroller, higher the number of pins necessary. (Shown in green and numbered starting from 2 and going up to 13) (Shown in light blue and numbered 0 and 1, if not used for serial data transmission)	No. of Analog Devices (such as sensors connected directly): Most sensors directly give analog values. These can be converted to digital values by using external circuitry, but if your applications demand analog input or you prefer using analog input, then remember that only a limited number of pins (around 4-12) accept analog input pins. Connecting any more analog inputs will need more pins. (Shown in blue and numbered from 0 to 5)
No. of Serial Ports: More the number of devices the board communicates with, higher the number of serial ports and pins required. (The serial ports are not separately available, they're actually digital pins that act as both, serial port and digital pins for connecting other devices also.) (Shown in light blue and numbered 0 and 1, if not used as digital data pins)	No. of PWM Pins: If you need voltage control using PWM, many more pins are required. (Just like the serial port pins, the PWM pins also perform dual functionality by moon-lighting as digital pins.) (Shown in green and numbered 9 to 11)

Analog out

Though most applications can use the digital form of analog signal called PWM, some applications (such as audio) might need exact analog output. In such case, the only available option is the Arduino™ Due, which has an ARM processor with analog output via a 12-bit digital to analog converter.